

Rates of Change & Concavity



Average rate of change

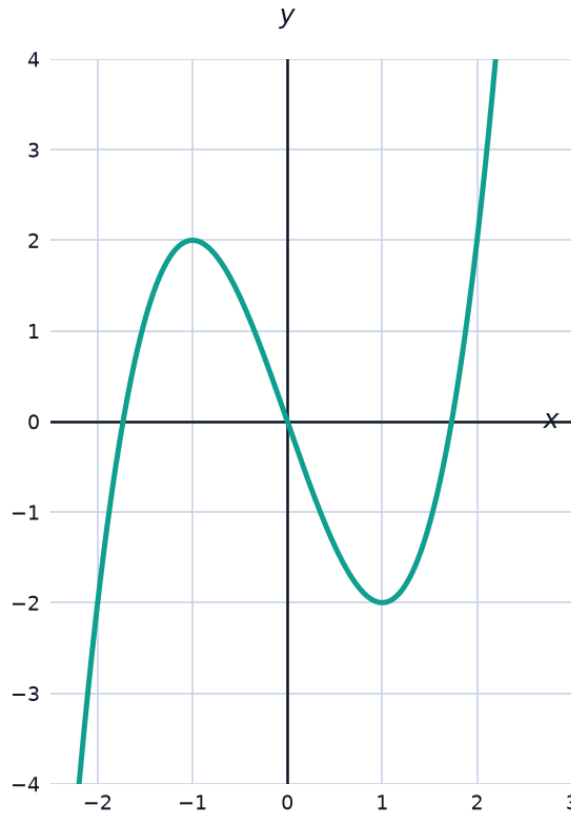
- 1 Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.
 - 2 A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.
-

Increasing/decreasing behavior from rate of change

- 3 A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.
 - 4 A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.
-

Visualizing concavity and inflection

- 5 The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.



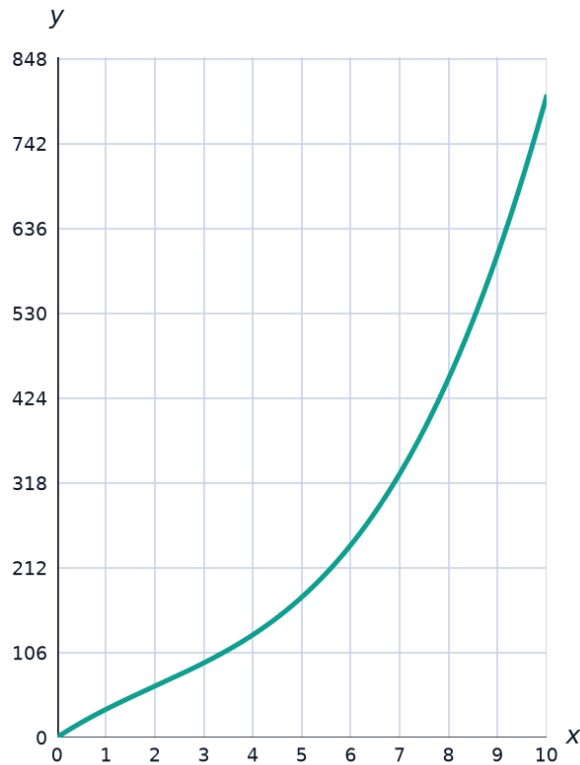
- a Describe the concavity of the graph on $(-2, 0)$.
- b Describe the concavity of the graph on $(0, 2)$.
- c State what kind of point $x = 0$ represents.

Rate of change of the rate of change

- 6 Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

Applying rate-of-change reasoning to models

- 7 A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \leq t \leq 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.



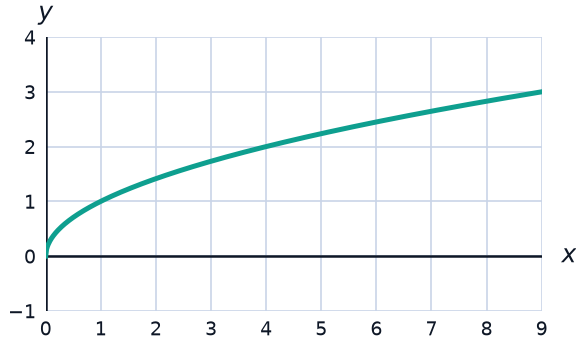
- 8 For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

Estimating rate of change from a table

- 9 A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.
- 10 Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

Visualizing rate of change on a graph

- 11 The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.



- a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$.
- b Explain how the shrinking steepness in the graph relates to your two AROC values.
- c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern.

Concavity and real-world interpretation

- 12 A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

Multiple choice — easy 15

- 13 Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.
- a. 35
 - b. -21
 - c. 33
 - d. 37
- 14 Find the average rate of change of $f(x) = 6x^2 - 4x$ on $[6, -6]$.
- a. -6
 - b. -4
 - c. -2
 - d. -76

15 Find the average rate of change of $f(x) = 9x^2 + 2x$ on $[5, -3]$.

- a. 18
- b. 20
- c. -70
- d. 22

16 Find the average rate of change of $f(x) = -5x^2 + 2x$ on $[5, 6]$.

- a. -53
- b. -55
- c. -3
- d. -51

17 Find the average rate of change of $f(x) = -9x^2 + x$ on $[4, 0]$.

- a. -35
- b. -33
- c. -37
- d. 37

18 Find the average rate of change of $f(x) = 7x^2 - 6x$ on $[-1, -2]$.

- a. -13
- b. -27
- c. -25
- d. -29

19 Find the average rate of change of $f(x) = 2x^2 - 5x$ on $[6, -3]$.

- a. -1
- b. -23
- c. 1
- d. 3

20 Find the average rate of change of $f(x) = 3x^2 - 4x$ on $[3, 5]$.

- a. 22
- b. 2
- c. 20
- d. 18

21 Find the average rate of change of $f(x) = -4x^2 + 9x$ on $[1, -2]$.

- a. 15
- b. 13
- c. 21
- d. 11

22 Find the average rate of change of $f(x) = 6x^2$ on $[3, -3]$.

- a. 0
- b. -36
- c. 2
- d. -2

- 23 Find the average rate of change of $f(x) = -3x^2$ on $[1, -4]$.
- a. 11
 - b. 9
 - c. 15
 - d. 7
- 24 Find the average rate of change of $f(x) = -9x^2 - 9x$ on $[-4, 0]$.
- a. -45
 - b. 25
 - c. 29
 - d. 27
- 25 Find the average rate of change of $f(x) = 9x^2 + 8x$ on $[4, -3]$.
- a. -55
 - b. 19
 - c. 17
 - d. 15
- 26 Find the average rate of change of $f(x) = x^2 + 4x$ on $[5, 4]$.
- a. 11
 - b. 15
 - c. 13
 - d. 3

- 27 Find the average rate of change of $f(x) = -6x^2 + x$ on $[-3, -2]$.
- a. 29
 - b. 33
 - c. -5
 - d. 31
-

Multiple choice — medium 15

- 28 Find the average rate of change of $f(x) = -2x^3$ on $[2, -5]$.
- a. -38
 - b. -42
 - c. -18
 - d. -34
- 29 Find the average rate of change of $f(x) = -2x^3$ on $[-1, 5]$.
- a. -42
 - b. -46
 - c. -32
 - d. -38
- 30 Find the average rate of change of $f(x) = 3x^3$ on $[2, -1]$.
- a. 15
 - b. 3
 - c. 11
 - d. 9

31 Find the average rate of change of $f(x) = -x^3$ on $[0, 5]$.

a. -21

b. -25

c. -27

d. -23

32 Find the average rate of change of $f(x) = -3x^3$ on $[4, 5]$.

a. -183

b. -189

c. -243

d. -177

33 Find the average rate of change of $f(x) = 3x^3$ on $[-2, 3]$.

a. 27

b. 15

c. 21

d. 3

34 Find the average rate of change of $f(x) = 4x^3$ on $[0, -3]$.

a. 40

b. 36

c. 44

d. 28

35 Find the average rate of change of $f(x) = 2x^3$ on $[-4, 1]$.

a. 30

b. 22

c. 26

d. 18

36 Find the average rate of change of $f(x) = 4x^3$ on $[2, -4]$.

a. 16

b. 48

c. 40

d. 56

37 Find the average rate of change of $f(x) = -4x^3$ on $[0, 2]$.

a. -24

b. -16

c. -8

d. -12

38 Find the average rate of change of $f(x) = -5x^3$ on $[-2, 3]$.

a. -25

b. -5

c. -35

d. -45

39 Find the average rate of change of $f(x) = -5x^3$ on $[1, -2]$.

- a. -15
- b. -19
- c. -5
- d. -25

40 Find the average rate of change of $f(x) = -5x^3$ on $[0, -3]$.

- a. -47
- b. -35
- c. -55
- d. -45

41 Find the average rate of change of $f(x) = -x^3$ on $[2, 3]$.

- a. -19
- b. -25
- c. -17
- d. -21

42 Find the average rate of change of $f(x) = 6x^3$ on $[3, -5]$.

- a. 24
- b. 102
- c. 126
- d. 114

Multiple choice — hard 15

- 43 A car's odometer reading (miles) is recorded every hour: $t = 0$: 131; $t = 1$: 150; $t = 2$: 178; $t = 3$: 215. Is the car's speed increasing, decreasing, or constant?
- a. Decreasing
 - b. Constant
 - c. Cannot be determined from this data
 - d. Increasing
- 44 A car's odometer reading (miles) is recorded every hour: $t = 0$: 75; $t = 1$: 111; $t = 2$: 138; $t = 3$: 156. Is the car's speed increasing, decreasing, or constant?
- a. Decreasing
 - b. Increasing
 - c. Constant
 - d. Cannot be determined from this data
- 45 A car's odometer reading (miles) is recorded every hour: $t = 0$: 62; $t = 1$: 81; $t = 2$: 104; $t = 3$: 136. Is the car's speed increasing, decreasing, or constant?
- a. Constant
 - b. Cannot be determined from this data
 - c. Decreasing
 - d. Increasing
- 46 A car's odometer reading (miles) is recorded every hour: $t = 0$: 133; $t = 1$: 148; $t = 2$: 168; $t = 3$: 194. Is the car's speed increasing, decreasing, or constant?
- a. Cannot be determined from this data
 - b. Decreasing
 - c. Increasing
 - d. Constant

- 47 A car's odometer reading (miles) is recorded every hour: $t = 0$: 151; $t = 1$: 174; $t = 2$: 197; $t = 3$: 220. Is the car's speed increasing, decreasing, or constant?
- a. Increasing
 - b. Constant
 - c. Decreasing
 - d. Cannot be determined from this data
- 48 A function h has average rates of change of 4 on $[0, 1]$, 2 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Concave up
 - b. Cannot be determined from this data
 - c. Linear (neither concave up nor down)
 - d. Concave down
- 49 A function h has average rates of change of 4 on $[0, 1]$, 7 on $[1, 2]$, and 12 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Linear (neither concave up nor down)
 - b. Cannot be determined from this data
 - c. Concave down
 - d. Concave up
- 50 A function h has average rates of change of 5 on $[0, 1]$, 7 on $[1, 2]$, and 11 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Linear (neither concave up nor down)
 - b. Concave down
 - c. Concave up
 - d. Cannot be determined from this data

- 51 A function h has average rates of change of 9 on $[0, 1]$, 8 on $[1, 2]$, and 6 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Cannot be determined from this data
 - b. Linear (neither concave up nor down)
 - c. Concave down
 - d. Concave up
- 52 A function h has average rates of change of 5 on $[0, 1]$, 0 on $[1, 2]$, and -1 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Concave down
 - b. Linear (neither concave up nor down)
 - c. Cannot be determined from this data
 - d. Concave up
- 53 Find the average rate of change of $f(x) = 6\sqrt{x}$ on $[4, 25]$.
- a. $\frac{13}{7}$
 - b. $\frac{6}{7}$
 - c. 2
 - d. $\frac{1}{7}$
- 54 Find the average rate of change of $f(x) = 9\sqrt{x}$ on $[1, 9]$.
- a. $\frac{13}{4}$
 - b. $\frac{9}{4}$
 - c. $\frac{9}{2}$
 - d. $\frac{1}{4}$

55 Find the average rate of change of $f(x) = 7\sqrt{x}$ on $[9, 49]$.

a. $\frac{7}{10}$

b. $\frac{7}{4}$

c. $\frac{1}{10}$

d. $\frac{17}{10}$

56 Find the average rate of change of $f(x) = 5\sqrt{x}$ on $[16, 49]$.

a. $\frac{5}{11}$

b. $\frac{1}{11}$

c. $\frac{16}{11}$

d. $\frac{5}{3}$

57 Find the average rate of change of $f(x) = 4\sqrt{x}$ on $[4, 36]$.

a. 1

b. $\frac{1}{8}$

c. $\frac{1}{2}$

d. $\frac{3}{2}$